The auxiliary torus of 7+1 spacetime: symbols and their definitions

A dictionary of every symbol used in the document on the auxiliary torus, with its defining equation and its name in the notebook 7+1.nb

Repository: github.com/once-ere/7plus1 — prepared for Nash
Notebook: 7+1.nb (Wolfram Language 15.0.1)
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Abstract

For each symbol this dictionary gives the mathematical notation, the name used in the Mathematica notebook 7+1.nb (where there is one; the names of section 25 begin with `tb`, those of section 26 with `rd`; "—" means that the notebook has no separate name), the defining equation and its meaning, grouped by topic in the order of the document. Longer defining equations follow each table as numbered displayed formulas. Section 13 lists the symbols that look alike, section 14 every Mathematica name of section 25. Every equation needed to define a symbol is written in this document.

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1. Conventions

- Units: $\hbar = c = 1$; energies, masses and inverse lengths in GeV; $\hbar c = 1.973269804 \times 10^{-16}$ GeV m (Mathematica `tbHbarCm`), so a length of 1 GeV^{-1} is $1.973269804 \times 10^{-16}$ m.
- Coordinates $X = (x_0, x_1, \ldots, x_7)$ (Mathematica `X = {x0, …, x7}`): x_0 is time (the direction of the octonion unit $e_0 = 1$), x_1, x_2, x_3 ordinary space, $x_4, \ldots, x_7$ the four auxiliary space-like coordinates.
- Indices: $A, B = 0, \ldots, 7$ (all directions); $\mu, \nu = 0, \ldots, 3$ (the directions of the world brane); $k, l = 4, \ldots, 7$ (the directions of the torus); $i, j = 1, 2, 3$ (ordinary space). A prime is d/dx_0, a dot d/dt; ln is the natural logarithm.

- Status labels used in the document: an ESTIMATE is an order of magnitude with numerical factors omitted; everything else is an exact consequence of the definitions, checked by section 25 of 7+1.nb and by tools/ref_torus_brane.py.

The signature: $\{7,1,0\}$ and $\mathrm{diag}(+1,-1,\ldots,-1)$. The 7+1 spacetime has one time-like dimension (x_0, the direction of the octonion unit $e_0 = 1$) and seven space-like dimensions (x_1, x_2, x_3 ordinary space, $x_4, \ldots, x_7$ auxiliary space). Two different objects describe this, and neither may be replaced by the other. (i) The tensor package xAct, used in section 12 of 7+1.nb exactly as in the author's notebook Einstein-Lovelock-7+1-Nash.nb, declares the metric with `DefMetric[{7, 1, 0}, g71[-a, -b], CD, …]`: the list $\{7,1,0\}$ declares seven space-like dimensions, one time-like dimension and no degenerate dimension — the numbers of entries $+1$ and -1 of $\eta_{71} = \mathrm{diag}(-1,+1,\ldots,+1)$ (`η71`), the same spacetime written in the sign convention of 4+4.nb (space-like = $+1$, time-like = -1). (ii) The matrix $\eta = -\eta_{71} = \mathrm{diag}(+1,-1,\ldots,-1)$ (`η17`) is simply the squares of the octonion basis: $e_0^2 = +1$ for the unit $e_0 = 1$ (time) and $e_i^2 = -1$ for the seven imaginary units $e_1, \ldots, e_7$ (space); more generally it is the real part of the product of two basis units, and it gives the quadratic form of the octonions:

$$\eta_{AB} = \mathrm{Re}(e_A e_B), \qquad e_0^2 = +1, \qquad e_i^2 = -1 \quad (i = 1,\ldots,7),$$
$$\mathrm{Re}(x\,x) = x_0^2 - x_1^2 - \cdots - x_7^2 = \eta_{AB}\, x^A x^B \qquad (x = x^A e_A \in \mathbb{O}).$$

η gives the signs of the Clifford relation $\Gamma_A\Gamma_B + \Gamma_B\Gamma_A = 2\,\eta_{AB}\,1_{16}$ of the real 16×16 Dirac matrices: time squares to $+1_{16}$ and each of the seven space directions to -1_{16}, exactly like the octonion units (in the notation $Cl(p,q)$, with p generators squaring to $+1$ and q to -1, the algebra $Cl(1,7) \cong \mathbb{R}(16)$). It also gives the signs of the metric components: $g_{00} > 0$ along the time-like direction, negative squares along the space-like directions. Every component computed by 7+1.nb follows from the component matrix of the metric; the declared list fixes for xAct the sign of the determinant of the metric, `SignDetOfMetric[g71]` $= -1$, which is the same in both sign conventions, since $\det\eta = \det\eta_{71} = -1$. 7+1.nb checks `SignatureOfMetric[g71]` $= \{7,1,0\}$, `SignDetOfMetric[g71]` $= -1$, $\mathrm{Re}(e_A e_B) = \eta_{AB}$ and $\mathrm{Re}(x\,x) = \eta_{AB}x^A x^B$.

2. Coordinates, the bulk and the torus

symbol	Mathematica	definition	meaning
$\vec{x}$	`{x1, x2, x3}`	(x_1, x_2, x_3)	ordinary space
$\vec{y}$, y_k	`tbY = {x4, x5, x6, x7}`	$\vec{y} = (x_4, x_5, x_6, x_7)$, $y_k = x_k$	the four auxiliary coordinates, coordinates on the torus
L_c	`tbLc`	(2.1)	coordinate period of each auxiliary coordinate
n_k, $\vec{n}$	—	$n_k \in \mathbb{Z}$, $\vec{n} \in \mathbb{Z}^4$	integers: lattice vectors, mode numbers
Λ (lattice)	—	(2.1)	the lattice $L_c\mathbb{Z}^4$ of identified translations
T^4	—	(2.1)	the auxiliary torus
S^1	—	$\mathbb{R}/L_c\mathbb{Z}$	a circle of period L_c; $T^4 = (S^1)^4$
T^2	—	$\mathbb{R}^2/(L_c\mathbb{Z})^2$	the two-torus (Example 3.2)
F	—	(2.2)	fundamental domain: one representative of each point of T^4
$\vec{e}_k$	—	$(\vec{e}_k)_l = \delta_{kl}$	unit vector along y_k
$f(\vec{y})$	`tbF` (Example 4.2)	(2.2)	a function on the torus (periodic on $\mathbb{R}^4$)
M	—	(2.3)	the bulk: the spacetime of the model
W	`tbOnW` (the rule $\vec{y} \to 0$)	(2.3)	the world brane, through $\vec{y} = 0$
g_{AB}, ds^2	`gCos` (cosmological metric)	$ds^2 = g_{AB}dX^A dX^B$; (2.4)	metric of the bulk
g_{kl}	`tbG4` (block of `gCos`)	$k, l = 4, \ldots, 7$	torus block of the metric (negative definite)

symbol	Mathematica	definition	meaning
$h_{\mu\nu}$	`tbInduced[g]`	$g_{\mu\nu}(x, \vec{y}=0)$	induced metric of the world brane
b, b_0	`b4[x0]`; `tbB` (symbol)	(2.4)	size (scale factor) of the torus; its value today
a	`a3[x0]`	(2.4)	scale factor of ordinary space
n (lapse)	`nP` (inhomogeneous metric of section 11.3)	(2.4)	the lapse; $n=1$ in cosmic time
dV_T	—	(2.5)	proper volume element of the torus
C	—	(2.5)	proper circumference of each circle
$V_\perp$	`tbV` (symbolic), `rIIVp` (number)	(2.5)	proper volume of the torus
$\ell_\perp$	`tbLp`	(2.5)	proper size of the torus
$d(\vec{y},\vec{y}\,')$	—	(2.6)	proper distance on the torus
λ, $\vec{y}\,'$, L_c', b'	`tbLam`, `tbDy` (with `tbd4` … `tbd7`)	(2.7)	rescaling of the coordinates; rescaled period and size

$$x_k \simeq x_k + L_c n_k, \quad T^4 = \mathbb{R}^4/\Lambda, \quad \Lambda = L_c\mathbb{Z}^4 = \{L_c\vec{n} : \vec{n} \in \mathbb{Z}^4\}, \quad \vec{y} \sim \vec{y}\,' \iff \vec{y} - \vec{y}\,' \in \Lambda, \tag{2.1}$$

$$F = [0, L_c)^4 \quad (\text{or } [-\tfrac{L_c}{2}, \tfrac{L_c}{2})^4), \qquad f(\vec{y} + L_c\vec{e}_k) = f(\vec{y}), \tag{2.2}$$

$$M = \mathbb{R}_{x_0} \times \mathbb{R}^3_{\vec{x}} \times T^4_{\vec{y}}, \qquad W = \{X \in M : x_4 = x_5 = x_6 = x_7 = 0\}, \tag{2.3}$$

$$ds^2 = n^2 dx_0^2 - a^2(dx_1^2 + dx_2^2 + dx_3^2) - b^2(dx_4^2 + \cdots + dx_7^2), \qquad b^8 = \det(-g_{kl}), \tag{2.4}$$

$$dV_T = \sqrt{\det(-g_{kl})}\, d^4y = b^4 d^4y, \qquad C = bL_c, \qquad V_\perp = \int_F b^4 d^4y = b^4 L_c^4, \qquad \ell_\perp = V_\perp^{1/4} = bL_c, \tag{2.5}$$

$$d(\vec{y},\vec{y}\,') = b \min_{\vec{n}\in\mathbb{Z}^4} |\vec{y} - \vec{y}\,' - L_c\vec{n}|, \tag{2.6}$$

$$\vec{y} = \lambda\vec{y}\,', \qquad L_c' = L_c/\lambda, \qquad b' = \lambda b, \qquad b'L_c' = bL_c, \qquad b'^4 L_c'^4 = b^4 L_c^4. \tag{2.7}$$

3. Symmetries of the torus

symbol	Mathematica	definition	meaning
$\vec{c}$	—	$\vec{y} \to \vec{y} + \vec{c}$	a translation of the torus ($\vec{c}$ modulo Λ)
$U(1)^4$	—	(3.1)	the group of translations
O	`tbSP` (all 384)	(3.2)	a signed permutation matrix
B_4	—	(3.2)	the hyperoctahedral group of the 384 signed permutations
shell k	`tbShell[k]`	$\{\vec{n} : \|\vec{n}\|^2 = k\}$ (with $\|n_i\| \le 3$ in the code)	the lattice vectors of a given length
orbit	`tbOrbits[k]`, `tbOrbitQ[k]`	(3.3)	vectors related by a signed permutation

$$U(1)^4 = \{\vec{y} \mapsto \vec{y} + \vec{c} \,(\text{mod } \Lambda)\}, \qquad \text{charge: the Kaluza–Klein momentum } \vec{n}, \tag{3.1}$$

$$O = \text{diag}(s_1, s_2, s_3, s_4)\, P_\pi, \quad s_i = \pm 1, \quad P_\pi \text{ a permutation matrix}; \quad |B_4| = 2^4 \cdot 4! = 384, \tag{3.2}$$

$$\vec{n} \simeq \vec{n}\,' \iff \vec{n}\,' = O\vec{n} \text{ for some } O \iff \{|n_i|\} = \{|n_i'|\} \text{ as multisets.} \tag{3.3}$$

4. Fields on the torus

symbol	Mathematica	definition	meaning
$\vec{q}$, $\vec{q}_{\vec{n}}$	`tbQ[n]`	(4.1)	wave vector in the coordinate $\vec{y}$ (physical momentum $\vec{q}/b$)
$\phi_{\vec{n}}$, $\phi^*_{\vec{n}}$	`tbPhi[n]`, `tbPhiC[n]`	(4.1)	normalized Fourier modes of the torus
$\int_F b^4 d^4y$	`tbInt[f]`	(4.2)	integral over the torus with the proper volume element
$c_{\vec{n}}$	`tbCoef[n]`	(4.2)	Fourier coefficients
$\delta_{\vec{n}\vec{m}}$	—	1 if $\vec{n}=\vec{m}$, else 0	Kronecker delta
$w(n_k)$	`tbW1[j]`	(4.3)	one-dimensional Fourier weights of Example 4.2
Δ_T	—	(4.4)	Laplacian of the torus
Φ	`tbWave` (one mode)	(4.5)	a scalar field of the bulk (Section 4.4 only)
$\Box$	`tbBox[f]`	(4.5)	wave operator (Laplace–Beltrami) of the bulk
m	`tbM`		mass of a bulk field (scalar or dirac16complex)
E, $\vec{k}$	`tbE`, `{tbk1, tbk2, tbk3}`	(4.5)	energy and momentum in ordinary space (Section 4.4)
$M_{\vec{n}}$	—	(4.6)	mass of the level $\vec{n}$ seen in ordinary spacetime
$\mu_{\vec{n}}$	—	(4.6)	Kaluza–Klein mass
zero mode	—	$\vec{n}=0$	the mode constant over the torus (mass m)
Kaluza–Klein tower	—	$\{M_{\vec{n}} : \vec{n}\in\mathbb{Z}^4\}$	all levels of one bulk field

$$\vec{q}_{\vec{n}} = \frac{2\pi\vec{n}}{L_c}, \qquad \phi_{\vec{n}}(\vec{y}) = \frac{e^{i\vec{q}_{\vec{n}}\cdot\vec{y}}}{\sqrt{V_\perp}}, \qquad \phi^*_{\vec{n}} = \frac{e^{-i\vec{q}_{\vec{n}}\cdot\vec{y}}}{\sqrt{V_\perp}}, \tag{4.1}$$

$$\int_F b^4 d^4y\, \phi^*_{\vec{n}}\phi_{\vec{m}} = \delta_{\vec{n}\vec{m}}, \qquad f = \sum_{\vec{n}} c_{\vec{n}}\phi_{\vec{n}}, \tag{4.2}$$

$$c_{\vec{n}} = \int_F b^4 d^4y\, \phi^*_{\vec{n}} f, \qquad \int_F b^4 d^4y\, |f|^2 = \sum_{\vec{n}} |c_{\vec{n}}|^2 \quad \text{(Parseval)},$$

$$f = \prod_{k=4}^{7} \cos^2\frac{\pi y_k}{L_c}, \qquad c_{\vec{n}} = \sqrt{V_\perp}\prod_k w(n_k), \tag{4.3}$$

$$w(0) = \tfrac{1}{2}, \qquad w(\pm1) = \tfrac{1}{4}, \qquad w(n) = 0 \quad (|n|\ge 2),$$

$$\Delta_T = \frac{1}{b^2}\sum_{k=4}^{7}\frac{\partial^2}{\partial y_k^2}, \qquad -\Delta_T\phi_{\vec{n}} = \frac{|\vec{q}_{\vec{n}}|^2}{b^2}\phi_{\vec{n}}, \tag{4.4}$$

$$\Box\Phi = \frac{1}{\sqrt{|g|}}\partial_A\big(\sqrt{|g|}\, g^{AB}\partial_B\Phi\big), \qquad (\Box + m^2)\Phi = 0, \tag{4.5}$$

$$\Phi = e^{-iEx_0 + i\vec{k}\cdot\vec{x}}\phi_{\vec{n}}(\vec{y}),$$

$$E^2 = |\vec{k}|^2 + M_{\vec{n}}^2, \qquad M_{\vec{n}} = \sqrt{m^2+\mu_{\vec{n}}^2}, \qquad \mu_{\vec{n}} = \frac{|\vec{q}_{\vec{n}}|}{b_0} = \frac{2\pi|\vec{n}|}{\ell_\perp}. \tag{4.6}$$

5. dirac16complex and the graviton on the torus

symbol	Mathematica	definition	meaning
ψ	`dirac16complex`		the 16-component complex bulk field
Γ_A, Γ^A	`Γ17[A]`, `Γ17up[A]`	(5.1)	real 16×16 Dirac matrices of 7+1, raised index
σ_i	`PauliMatrix[i]`	(5.2)	Pauli matrices
γ_μ, γ_5	`g4D[[μ+1]]`, `g5D`	(5.2)	4×4 Dirac matrices of the brane (Dirac representation)
ρ_k	`rhoK[[k-3]]`	(5.3)	four anti-Hermitian 4×4 matrices for the transverse directions
R_A	`rBr[[A+1]]`	(5.3)	the brane-adapted 16×16 representation
U	`uInt`	(5.4)	unitary intertwiner, $\Gamma_A U = U R_A$
ζ, a	—	(5.4)	the field in the brane-adapted basis; the transverse index $a=1,\ldots,4$
${e_A}^B$	—	(5.5)	frame (vielbein) of the static torus
$\vec{p}$, E	—	(5.5)	momentum and energy in ordinary space
$\gamma\cdot p$	—	(5.5)	the brane Dirac operator in momentum space
X (operator)	`xKK` (section 18.1)	(5.6)	the Kaluza–Klein operator of dirac16complex
D	—	$D=8$	number of spacetime dimensions
$g_{\mu\nu}$, $g_{\mu k}$, g_{kl}	—	(5.7)	graviton components: 4D graviton, graviphotons, moduli

$$\Gamma_A\Gamma_B+\Gamma_B\Gamma_A=2\eta_{AB}1_{16},\qquad \Gamma^A=\eta^{AA}\Gamma_A\quad\text{(no sum)},\tag{5.1}$$

$$\sigma_1=\begin{pmatrix}0&1\\1&0\end{pmatrix},\qquad \sigma_2=\begin{pmatrix}0&-i\\i&0\end{pmatrix},\qquad \sigma_3=\begin{pmatrix}1&0\\0&-1\end{pmatrix},\tag{5.2}$$

$$\gamma_0=\mathrm{diag}(1,1,-1,-1),\qquad \gamma_i=\begin{pmatrix}0&\sigma_i\\-\sigma_i&0\end{pmatrix},\qquad \gamma_5=i\gamma_0\gamma_1\gamma_2\gamma_3,$$

$$\rho_4=i\gamma_0,\quad \rho_5=-\gamma_1,\quad \rho_6=-\gamma_2,\quad \rho_7=-\gamma_3,\qquad \rho_k\rho_l+\rho_l\rho_k=-2\delta_{kl}1_4,\tag{5.3}$$

$$R_\mu=\gamma_\mu\otimes1_4\quad(\mu=0,\ldots,3),\qquad R_k=\gamma_5\otimes\rho_k\quad(k=4,\ldots,7),$$

$$\Gamma_A U=UR_A\quad(A=0,\ldots,7),\qquad U^\dagger U=1_{16},\tag{5.4}$$

$$\psi=U\zeta,\qquad \zeta=\sum_{a=1}^{4}\psi_a\otimes e_a\in\mathbb{C}^4\otimes\mathbb{C}^4,$$

$${e_A}^B=\mathrm{diag}(1,1,1,1,b_0^{-1},b_0^{-1},b_0^{-1},b_0^{-1}),\qquad (i\Gamma^A{e_A}^B\partial_B-m)\psi=0,\tag{5.5}$$

$$\gamma\cdot p=E\gamma_0-\sum_{i=1}^{3}p_i\gamma_i,$$

$$X=\gamma_5\otimes\frac{\vec\rho\cdot\vec q}{b_0},\qquad X^2=-\frac{|\vec q|^2}{b_0^2}1_{16},\tag{5.6}$$

$$(\gamma\cdot p\otimes1_4+X-m)\zeta=0\quad\Rightarrow\quad E^2=|\vec p|^2+m^2+\frac{|\vec q|^2}{b_0^2},$$

$$\frac{D(D-3)}{2} = 20 = \underbrace{2}_{g_{\mu\nu}} + \underbrace{4\times 2}_{g_{\mu k}} + \underbrace{10}_{g_{kl}} \ \text{(zero mode)}, \qquad (5.7)$$
$$20 = 5 + 3\times 3 + 6 \ \text{(each massive level)}.$$

In (5.4) e_a ($a = 1, \ldots, 4$) are the unit vectors of $\mathbb{C}^4$ and ψ_a the four brane spinors. Fermions are PERIODIC around all four circles in 7+1.nb (one of the $2^4 = 16$ spin structures of T^4); an antiperiodic circle would shift its momenta to $2\pi(n_k + \frac{1}{2})/L_c$.

6. Counting the levels

symbol	Mathematica	definition	meaning
k (shell)	—	$k = \lvert\vec{n}\rvert^2$	the squared length of a lattice vector
$r_4(k)$	`rIIr4[[k]]` (section 22.2)	(6.1)	number of $\vec{n} \in \mathbb{Z}^4$ with $\lvert\vec{n}\rvert^2 = k$
$\sigma(k)$, d	`rIIsig`	(6.1)	divisor sum; a divisor of k
$k_{\max}$	`rIIkmax`	$\lfloor (m_r \ell_\perp/(4\pi))^2 \rfloor$ ($m_r = 10$ GeV)	largest shell with $\mu < m_r/2$ (the radion decays of section 22.2 of 7+1.nb)
$N(R)$, R	`tbNR[R]`, `tbRs`, `tbNRs`	(6.2)	number of lattice vectors in the ball of radius R
Weyl deviation	`tbDev`	$N(R)/(\pi^2 R^4/2) - 1$	relative deviation from Weyl's law (6.2)
$N(\mu)$, dN	`tbNmu[mu]`	(6.3)	number of levels below μ; their density
$F(\mu)$	—	(6.3)	any smooth function of the level mass
u	—	$u = \mu/m_h$	integration variable (Example 5.3)

$$r_4(k) = 8\sigma(k) - 32\sigma(k/4) \ (k \geq 1), \quad r_4(0) = 1, \quad \sigma(k) = \sum_{d|k} d, \quad \sigma(k/4) = 0 \text{ unless } 4 \mid k, \qquad (6.1)$$

$$N(R) = \#\{\vec{n} \in \mathbb{Z}^4 : \lvert\vec{n}\rvert \leq R\} = \sum_{k=0}^{R^2} r_4(k) \simeq \frac{\pi^2}{2} R^4, \qquad (6.2)$$

$$N(\mu) \simeq \frac{V_\perp \mu^4}{32\pi^2}, \qquad dN = \frac{V_\perp \mu^3 d\mu}{8\pi^2}, \qquad \sum_{\vec{n}} F(\mu_{\vec{n}}) \simeq \frac{V_\perp}{8\pi^2} \int_0^\infty \mu^3 F(\mu)\, d\mu. \qquad (6.3)$$

7. Gravity

symbol	Mathematica	definition	meaning
κ, G_8, M_8	`tbKap` (symbol), `rIIM8`	(7.1)	gravitational coupling, Newton constant and reduced Planck mass of eight dimensions
G_4, $M_{\rm Pl}$	`tbGN` (symbol), `tbMP`, `rIIMPl`	(7.1)	Newton constant and reduced Planck mass of four dimensions
S_1	—	(7.1)	Einstein–Hilbert action of the bulk
R (curvature), $R^{(4)}$, $g^{(4)}$	`cvCos["R"]`	(7.2)	scalar curvature of the bulk; of the four-dimensional metric $g^{(4)}$

symbol	Mathematica	definition	meaning
H_a, H_b	`Ha`, `Hb`	(7.2)	expansion rates of ordinary and auxiliary space
Φ_{N}, Φ_8	—	(7.3)	Newtonian potential; that of a point mass in infinite eight-dimensional space
G_D, d, ρ	—	(7.3)	Newton constant in D dimensions; $d = D-1$ space dimensions; mass density
Ω_{d-1}, $\Gamma(z)$	`Gamma`	(7.3)	area of the unit sphere in d dimensions; Euler's Gamma function
M, M', r	`tbr` (symbol)		two masses on the brane; their distance along the brane
$S(r)$	`tbS[r]`	(7.4)	image sum (lengths in units of $\ell_\perp$)
$\hat{f}(\vec{p})$, p	—	(7.5)	Fourier transform on $\mathbb{R}^4$; $p = \lvert\vec{p}\rvert$
J_ν, K_ν	`BesselJ`, `BesselK`	(7.5)	Bessel and modified Bessel functions
Kaluza–Klein side	`tbSkk[r]`, `tbPois`	(7.6)	the right side of the Poisson identity; the table of both sides
$\mu_{\vec{m}}$	—	$2\pi\lvert\vec{m}\rvert/\ell_\perp$	Kaluza–Klein mass of the dual vector $\vec{m}$
$Z(5/2)$	`tbZ52`, `tbZ52s`	(7.7)	Epstein zeta value of the lattice (closed form; shell sum)
$\zeta(s)$	`Zeta`	$\sum_{k\ge1} k^{-s}$	Riemann zeta function
near/far ratios	`tbNear`, `tbFar`	(7.7)	checks of the limits $r \ll \ell_\perp$, $r \gg \ell_\perp$
β, ϕ	`betR` (symbol, section 20.3)	(7.8)	the radion and the canonical radion
$U(b)$, U_0, K_U	`uPot`, `u0B`, `kU`	(7.8)	stabilizing potential, its minimum value, its stiffness
m_r	`rIIm` (units $\ell_2 = 1$)	(7.8)	radion mass (10 GeV in sections 22.2–24)
ε, p_a, p_b	`feStress[W]` (section 20.2)	(7.9)	energy density and the pressures along ordinary and auxiliary space
Λ (cosmological)	`lamb`	(7.9)	bulk cosmological constant
g_E	—	(7.10)	four-dimensional Einstein-frame metric
$T^\mu{}_\mu$, $\mathcal{L}_{\phi W}$, V_ϕ	—	(7.10)	trace of the brane energy–momentum tensor; radion coupling; radion potential

$$S_1 = -\frac{1}{2\kappa}\int d^8X\sqrt{|g|}\,R, \qquad \kappa = 8\pi G_8 = M_8^{-6}, \tag{7.1}$$

$$8\pi G_4 = \frac{\kappa}{V_\perp} = \frac{1}{M_{\mathrm{Pl}}^2}, \qquad M_{\mathrm{Pl}}^2 = V_\perp M_8^6,$$

$$R = -2(3H_a' + 6H_a^2 + 4H_b' + 10H_b^2 + 12H_aH_b), \qquad H_a = \frac{a'}{a}, \quad H_b = \frac{b'}{b}, \tag{7.2}$$

$$R^{(4)} = -6(H_a' + 2H_a^2) \quad \text{(the value of } R \text{ for } H_b = 0),$$

$$\nabla^2\Phi_{\mathrm{N}} = 8\pi G_D\frac{D-3}{D-2}\rho, \qquad g_{00} = 1 + 2\Phi_{\mathrm{N}}, \qquad \Omega_{d-1} = \frac{2\pi^{d/2}}{\Gamma(d/2)}, \tag{7.3}$$

$$\Phi_8(r) = -\frac{5\kappa M}{32\pi^3 r^5},$$

$$\Phi_{\rm N}(r) = -\frac{5\kappa M}{32\pi^3}\sum_{\vec n}\frac{1}{(r^2+\ell_\perp^2|\vec n|^2)^{5/2}}, \qquad S(r) = \sum_{\vec n\in\mathbb{Z}^4}\frac{1}{(r^2+|\vec n|^2)^{5/2}}, \tag{7.4}$$

$$\hat f(\vec p) = \int d^4y\, e^{-i\vec p\cdot\vec y} f(\vec y), \qquad \sum_{\vec n} f(\vec n) = \sum_{\vec m}\hat f(2\pi\vec m), \tag{7.5}$$

$$\int d^4y\,\frac{e^{-i\vec p\cdot\vec y}}{(r^2+|\vec y|^2)^{5/2}} = \frac{4\pi^2}{3r}e^{-pr},$$

$$S(r) = \frac{4\pi^2}{3r}\sum_{\vec m\in\mathbb{Z}^4} e^{-2\pi r|\vec m|}, \qquad \Phi_{\rm N} = -\frac{5}{3}\frac{G_4 M}{r}\sum_{\vec m} e^{-\mu_{\vec m} r}, \tag{7.6}$$

$$Z(5/2) = \sum_{\vec n\neq 0}|\vec n|^{-5} = 7\zeta(\tfrac52)\zeta(\tfrac32), \qquad S(r) = r^{-5} + Z(5/2) + O(r^2)\ \ (r\to 0), \tag{7.7}$$

$$\frac{3r}{4\pi^2}S(r) - 1 \simeq 8e^{-2\pi r}\ \ (r\to\infty),$$

$$U(b) = U_0 + \frac{K_U}{2}\ln^2\frac{b}{b_0}, \qquad \beta = \ln\frac{b}{b_0}, \qquad m_r^2 = \frac{\kappa K_U}{12}, \qquad \phi = \sqrt{12}\,M_{\rm Pl}\beta, \tag{7.8}$$

$$\beta'' + 3H_a\beta' + 4\beta'^2 = \kappa\Big(\frac{\varepsilon}{6} - \frac{p_a}{2} + \frac{p_b}{3}\Big) + \frac{\Lambda}{3}, \tag{7.9}$$

$$h = (b_0/b)^4 g_E, \qquad \mathcal{L}_{\phi W} = \frac{\phi\, T^\mu{}_\mu}{\sqrt{3}M_{\rm Pl}}, \qquad V_\phi(r) = -\frac{2}{3}\frac{G_4 M M'}{r}e^{-m_r r}. \tag{7.10}$$

8. The life of the torus

symbol	Mathematica	definition	meaning
τ	—	$x_0 - x_s$	time since the singularity x_s
p, q	—	$a\propto\tau^p$, $b\propto\tau^q$	Kasner exponents
p_1, q_1; p_2, q_2	`rIp1`, `rIq1`, `rIp2`, `rIq2`	(8.1)	the Einstein and the Gauss–Bonnet Kasner exponents
α_2, α_3	`alf[2]`, `alf[3]`		Gauss–Bonnet and cubic Lovelock couplings
ℓ_2, ξ	—, `rIxi`	$\ell_2=\sqrt{\alpha_2}$, $\xi=\alpha_3/\alpha_2^2=10^{-6}$	unit of time of sections 21–22.1; the only vacuum parameter
h_0, h_*	`rIcub[[1]]`, `rIhStar`	(8.2)	limits of H_b in the cubic regime, without and with a gas
P, Q, w_a, w_b	—	(8.3)	power laws with a fluid; its equation-of-state ratios
A, ϑ, τ_E, H_E	`rIIaF`, `rIIphF`, `rIItEF`	(8.4)	fitted radion oscillation after the capture; Einstein-frame rate
$\beta_{\rm eq}$	—	(8.5)	displacement of the radion from b_0
Λ_4, f_W	`fWb` (symbol)	(8.5)	four-dimensional cosmological constant; brane tension
H_0, Ω_Λ, H_∞	`rIIH0g`, `rIIOL`, `rIIIH`	(8.6)	Hubble rate and dark-energy fraction today; final de Sitter rate
H (de Sitter)	—	(8.6)	expansion rate of eight-dimensional de Sitter space

$$p_1 = \frac{1+2\sqrt2}{7}, \quad q_1 = \frac{2-3\sqrt2}{14} \quad (3p+4q=1,\ 3p^2+4q^2=1); \qquad 3p_2+4q_2=3, \tag{8.1}$$

$$h_0 = -\sqrt{\alpha_2/(30\alpha_3)}, \qquad h_* = -\sqrt{\alpha_2/(54\alpha_3)}, \tag{8.2}$$

$$p_a = w_a\varepsilon, \quad p_b = w_b\varepsilon, \quad 3P(1+w_a) + 4Q(1+w_b) = 2; \quad (w_a, w_b) = (0, \tfrac{1}{4}) \Rightarrow (P, Q) = (0, \tfrac{2}{5}), \tag{8.3}$$

$$\beta \simeq \frac{A\sin(m_r\tau + \vartheta)}{m_r(\tau - \tau_E)}, \qquad H_E = H_a + 2H_b \simeq \frac{2}{3(\tau - \tau_E)}, \tag{8.4}$$

$$\beta_{\mathrm{eq}} = \frac{\Lambda_4 + \kappa f_W/V_\perp + \frac{\kappa}{2}(\varepsilon - 3p_a) + \kappa p_b - 12\alpha_2 H_a^2(H_a' + H_a^2)}{3m_r^2}, \quad \Lambda_4 = \Lambda + \kappa U_0 + \frac{\kappa f_W}{V_\perp}, \tag{8.5}$$

$$H_\infty = H_0\Omega_\Lambda^{1/2}, \qquad a = b = e^{Hx_0} : \; 21H^2 - 420\alpha_2 H^4 + 2520\alpha_3 H^6 = \Lambda. \tag{8.6}$$

9. Kaluza–Klein momentum, the world brane and the delta function

symbol	Mathematica	definition	meaning
$\vec{n}$, $\vec{m}$, $\vec{l}$	`tbVtx[n, m, l]`	(9.1)	mode numbers of a cubic vertex; the overlap of three modes
$\chi_{\vec{n}}(x)$	—	(9.2)	four-dimensional field of the level $\vec{n}$
$\delta^4(\vec{y})$, $\delta(y)$	`DiracDelta`	(9.3)	delta function of the torus; of one circle
n (integer)	`tbn`	$n \in \mathbb{Z}$	Fourier index in the one-dimensional check of (9.3)
ρ_4	`tbRho` (symbol)		energy per brane volume of a brane source
ε_8	—	(9.4)	its average eight-dimensional density (the smeared source)

$$\int_F b^4 d^4y\, \phi_{\vec{n}}\phi_{\vec{m}}\phi^*_{\vec{l}} = \frac{\delta_{\vec{n}+\vec{m},\vec{l}}}{\sqrt{V_\perp}}, \qquad (\phi_{\vec{n}}\phi_{\vec{m}}\phi^*_{\vec{l}})(0) = V_\perp^{-3/2}, \tag{9.1}$$

$$\psi(x,\vec{y}) = \sum_{\vec{n}} \chi_{\vec{n}}(x)\phi_{\vec{n}}(\vec{y}), \qquad \psi(x,0) = \frac{1}{\sqrt{V_\perp}}\sum_{\vec{n}}\chi_{\vec{n}}(x), \tag{9.2}$$

$$\delta^4(\vec{y}) = \delta(x_4)\delta(x_5)\delta(x_6)\delta(x_7) = \frac{1}{L_c^4}\sum_{\vec{n}} e^{i\vec{q}_{\vec{n}}\cdot\vec{y}}, \qquad \int_{-L_c/2}^{L_c/2} e^{-iq_n y}\delta(y)\,dy = 1, \tag{9.3}$$

$$\varepsilon_8 = \frac{1}{V_\perp}\int_F b^4 d^4y\, \frac{\rho_4\delta^4(\vec{y})}{b^4} = \frac{\rho_4}{V_\perp}. \tag{9.4}$$

10. The vacuum energy of the torus

symbol	Mathematica	definition	meaning
$\rho^{(1)}_{\mathrm{Cas}}$	`dkCas[mℓ]` (section 24.5)	(10.1)	Casimir energy per brane volume, one real bosonic degree of freedom
$Z(4)$	`dkZ48`	(10.1)	Epstein zeta value $\sum_{\vec{n}\neq 0}\lvert\vec{n}\rvert^{-8}$
ρ_Λ	`dkRhoL`	(10.2)	observed dark-energy density
G (lattice metric)	—	(10.3)	metric of a general lattice (section 11 of the document)

$$\rho^{(1)}_{\mathrm{Cas}} = -\frac{2}{\ell_\perp^4}\sum_{\vec{n}\neq 0}\left(\frac{m\ell_\perp}{2\pi|\vec{n}|}\right)^4 K_4(m\ell_\perp|\vec{n}|) \xrightarrow{m\to 0} -\frac{\Gamma(4)Z(4)}{\pi^4\ell_\perp^4}, \tag{10.1}$$

$$Z(4)=\sum_{\vec n\neq 0}\frac{1}{|\vec n|^8}=\sum_{k\geq 1}\frac{r_4(k)}{k^4}=8(1-4^{-3})\zeta(4)\zeta(3),$$

$$\rho_\Lambda = 3H_0^2M_{\rm Pl}^2\Omega_\Lambda, \tag{10.2}$$

$$\mu^2=\vec q\cdot G^{-1}\vec q\quad\text{(a general flat torus }\mathbb{R}^4/\Lambda,\ \vec q\text{ in the dual lattice).} \tag{10.3}$$

A fermion contributes to the Casimir energy with the opposite sign; dirac16complex has 16 real degrees of freedom on shell, the eight-dimensional graviton 20.

11. The 4-sphere of the comparison, and the radion

symbol	Mathematica	definition	meaning
S^4	—	(11.1)	the 4-sphere (comparison, section 10 of the document)
r, r_0	`rdR0` (r_0)	$r=r_0e^\beta$	radius of the sphere; its reference value
$\hat\gamma$	`rdGam4`	(11.1)	metric of the unit 4-sphere
γ	—	$b_0^2\delta$ (T^4), $r_0^2\hat\gamma$ (S^4)	reference metric of the extra space
β	`rdBt[x0]`, `rdBe`	$\ln(b/b_0)$ or $\ln(r/r_0)$	the radion
g_E	—	(11.2)	four-dimensional Einstein-frame metric
V_0	—	$b_0^4L_c^4$ or $\frac{8\pi^2}{3}r_0^4$	volume at $\beta=0$
$V(\beta)$, σ	`rdV`	(11.3)	radion potential; 0 for T^4, 1 for S^4
U, U_0, U_1, K_U	`rdU`, `rdU0`, `rdU1`, `rdKU`	(11.3)	stabilizing energy density and its expansion
Λ	`rdLam`	bulk cosmological constant	enters $V(\beta)$ as $M_{\rm Pl}^2\Lambda e^{-4\beta}$
l, d_l	`rdDl`	(11.4)	degree of a spherical harmonic; multiplicity
N, L, y	`rdNS4`	(11.4)	number of sphere states; largest degree; $y=L+2$
Z_l, $C_l^{(3/2)}$	`rdZon`	(11.5)	zonal harmonic; Gegenbauer polynomial
r_{S^4}	`rdRS4`	$(3V_\perp/(8\pi^2))^{1/4}$	radius of a sphere of volume $V_\perp$
λ (Dirac)	—	$\pm i(l+2)/r$	eigenvalues of the Dirac operator of S^4
$\tilde\gamma_a$	`rdGt`	$i\rho_{a+3}$	Hermitian Dirac matrices of S^4

$$S^4:\ g_{kl}=-r^2\hat\gamma_{kl},\quad \hat\gamma=\mathrm{diag}(1,\sin^2x_4,\sin^2x_4\sin^2x_5,\sin^2x_4\sin^2x_5\sin^2x_6),\quad V=\frac{8\pi^2}{3}r^4, \tag{11.1}$$

$$ds^2=e^{-4\beta}g_{E\mu\nu}dx^\mu dx^\nu-e^{2\beta}\gamma_{kl}dy^kdy^l,\qquad \phi=\sqrt{12}M_{\rm Pl}\beta,\quad M_{\rm Pl}^2=\frac{V_0}{\kappa}, \tag{11.2}$$

$$V(\beta)=M_{\rm Pl}^2\Big[e^{-4\beta}\big(\Lambda+\kappa U(\beta)\big)-\frac{6\sigma}{r_0^2}e^{-6\beta}\Big],\quad U=U_0+U_1\beta+\frac{K_U}{2}\beta^2,\quad m_r^2=\frac{V''(0)}{12M_{\rm Pl}^2}, \tag{11.3}$$

$$\mu_l=\frac{\sqrt{l(l+3)}}{r},\quad d_l=\frac{(2l+3)(l+1)(l+2)}{6},\quad N=\sum_{l\leq L}d_l=\frac{y^2(y^2-1)}{12}, \tag{11.4}$$

$$Z_l=\sqrt{\frac{d_l}{V}}\,\frac{C_l^{(3/2)}(\cos x_4)}{C_l^{(3/2)}(1)},\quad C_l^{(3/2)}(1)=\frac{(l+1)(l+2)}{2}. \tag{11.5}$$

12. Constants and data

symbol	Mathematica	value	meaning and source
M_8	`rIIM8`	10^4 GeV	reduced eight-dimensional Planck mass (choice of sections 22–24)
$M_{\rm Pl}$	`rIIMPl`	2.435×10^{18} GeV	reduced Planck mass
m_r	—	10 GeV	radion mass (choice of sections 22–24)
m_h	`mHb`	125.20 GeV	Higgs mass (PDG 2025)
r_p	—	0.8414 fm	proton charge radius (CODATA 2018)
a_0	—	$5.29177210903 \times 10^{-11}$ m	Bohr radius (CODATA 2018)
$\hbar c$	`tbHbarCm`	$1.973269804 \times 10^{-16}$ GeV m	CODATA 2018
$R_{\rm PDG}$	—	(12.1)	radius of each circle in the convention of the Particle Data Group
M_D	—	(12.1)	fundamental scale in that convention

$$V_\perp = (2\pi R_{\rm PDG})^4, \qquad M_{\rm Pl}^2 = R_{\rm PDG}^4 M_D^6 \quad \Rightarrow \quad R_{\rm PDG} = \frac{\ell_\perp}{2\pi}, \qquad M_D = (2\pi)^{2/3} M_8. \tag{12.1}$$

13. Symbols that look alike

- R: the scalar curvature R (and $R^{(4)}$); the radius R of a ball of lattice vectors in $N(R)$; the PDG radius $R_{\rm PDG}$; the brane-adapted Dirac matrices R_A (always with an index).
- Λ: the lattice $\Lambda = L_c\mathbb{Z}^4$ (sections 3, 11) and the bulk cosmological constant Λ (sections 6.5, 7, 10); Λ_4 is the four-dimensional cosmological constant.
- Φ: the scalar field of section 4.4; $\Phi_{\rm N}$ and Φ_8 are gravitational potentials (in the rest of 7+1.nb Φ is the Higgs doublet).
- ϕ: the Fourier modes $\phi_{\vec{n}}$ (always with an index) and the canonical radion ϕ.
- ζ: the spinor ζ of section 4.5 and the Riemann zeta function $\zeta(s)$ (always with an argument); $Z(5/2)$, $Z(4)$ are lattice sums.
- k: the shell $k = |\vec{n}|^2$; the index $k = 4, \ldots, 7$ of the torus directions; the momentum $\vec{k}$ in ordinary space (section 4.4).
- K: the Bessel functions K_ν; the stiffness K_U of the stabilizing potential.
- M: the bulk M; a mass M (and M'); the level mass $M_{\vec{n}}$; M_8, $M_{\rm Pl}$, M_D.
- μ: the index $\mu = 0, \ldots, 3$; the Kaluza–Klein mass $\mu_{\vec{n}}$ (with an index); a generic mass μ in $N(\mu)$.
- e: the octonion units e_A; the unit vectors $\vec{e}_k$ of $\mathbb{R}^4$ and e_a of $\mathbb{C}^4$; the frame ${e_A}^B$.
- S: the image sum $S(r)$; the action S_1; the circle S^1.
- Γ: the Dirac matrices Γ_A (with an index) and Euler's Gamma function $\Gamma(z)$ (with an argument).
- ρ: the densities ρ, ρ_4, ρ_Λ, $\rho_{\rm Cas}$, and the 4×4 matrices ρ_k ($k = 4, \ldots, 7$).
- σ: the divisor sum $\sigma(k)$ and the Pauli matrices σ_i.
- n: the lattice vector $\vec{n}$ and its integers n_k; the lapse n of (2.4).

- a: the scale factor a of ordinary space and the transverse spinor index $a = 1, \dots, 4$.
- d: a divisor d in $\sigma(k)$; the number d of space dimensions; the distance $d(\vec{y}, \vec{y}\,')$.
- F: the fundamental domain F and a generic function $F(\mu)$; f is a function on the torus, f_W the brane tension.
- p, q: the Kasner exponents p, q (and p_1, q_1, p_2, q_2); the momentum $\vec{p}$ and the Fourier variable p; the pressures p_a, p_b; the wave vector $\vec{q}$ (always with an arrow).
- w: the Fourier weights $w(n_k)$ and the equation-of-state ratios w_a, w_b.
- h: the induced metric $h_{\mu\nu}$ (with indices) and the cubic-regime rates h_0, h_*.
- H: the expansion rates H_a, H_b, H_E, H_0, H_∞ and the de Sitter rate H.
- X: the coordinates X and the Kaluza–Klein operator X of dirac16complex (section 4.5).
- U: the intertwiner U; the stabilizing potential $U(b)$ (with its argument) and U_0; the group $U(1)^4$.
- A: the index $A = 0, \dots, 7$ and the amplitude A of the radion oscillation.
- C: the circumference $C = bL_c$; the Gegenbauer polynomial $C_l^{(3/2)}$ (section 11).
- r: the distance r between two masses; the radius r of the 4-sphere of section 11 (and r_0, r_{S^4}); $r_4(k)$ the lattice multiplicity.
- λ: the rescaling λ of (2.7) and the Dirac eigenvalue λ on S^4 (section 11).
- γ: the brane Dirac matrices γ_μ, γ_5; the reference metric γ of the extra space and $\hat{\gamma}$ of the unit 4-sphere; the Hermitian $\tilde{\gamma}_a$ of S^4.
- σ: also the curvature switch σ of $V(\beta)$ in (11.3) (0 for T^4, 1 for S^4).
- Z: the lattice sums $Z(5/2)$, $Z(4)$ and the zonal harmonics Z_l of S^4.

14. Mathematica names

14.1 The names of section 25

name	meaning
`tbG4`, `tbDy`, `tbd4` … `tbd7`	torus block g_{kl} of `gCos`; the coordinate differentials $(dy_4, \dots, dy_7)$
`tbB`, `tbLc`, `tbLam`	symbols b (also b_0), L_c, λ
`tbSP`, `tbShell`, `tbOrbits`, `tbOrbitQ`	the 384 signed permutation matrices; the shell k (vectors with $\lvert n_i\rvert \le 3$); its orbits; the orbit test
`tbY`, `tbV`, `tbQ`, `tbPhi`, `tbPhiC`	$\vec{y}$; $V_\perp = b^4 L_c^4$; $\vec{q}_{\vec{n}}$; $\phi_{\vec{n}}$; $\phi^*_{\vec{n}}$
`tbInt`, `tbF`, `tbW1`, `tbCoef`	$\int_F b^4 d^4y$; the bump f; $w(n_k)$; $c_{\vec{n}}$
`tbGS`, `tbBox`, `tbWave`, `tbE`, `tbk1` … `tbk3`, `tbM`	static metric $\mathrm{diag}(1, -1, -1, -1, -b^2, \dots)$; $\Box$; one mode; E, $\vec{k}$, m
`tbNR`, `tbRs`, `tbNRs`, `tbDev`	$N(R)$; the radii 10, 30, 100, 300, 1000; their $N(R)$; deviations from Weyl's law
`tbLp`, `tbHbarCm`, `tbNmu`, `tbLevels`	$\ell_\perp$; $\hbar c$ in GeV m; $N(\mu)$; the table of the first levels
`tbKap`, `tbGN`, `tbr`, `tbMP`, `tbVs`	symbols κ, G, r, $M_{\rm Pl}$, $V_\perp$ of the Newton-law checks
`tbS`, `tbSkk`, `tbPois`	image sum $S(r)$ (shells to $k_{\max}$ and a continuum tail); Kaluza–Klein sum (shells to 20000); the table of both
`tbZ52`, `tbZ52s`, `tbNear`, `tbFar`	$Z(5/2)$ closed form and shell sum; the near- and far-field ratios
`tbVtx`, `tbn`, `tbRho`	overlap of three modes; an integer n; the symbol ρ_4
`tbJ`, `tbOnW`, `tbInduced`, `tbPt`, `tbChrW`	the Jacobian J; the rule $\vec{y} \to 0$; $h = J^{\mathsf{T}} g J$; the point $x_0 = \frac{1}{2}$, $x_4 = 0$; Christoffel symbols $\{k\,\mu\nu\}$ at the brane
`tbSPermQ`, `tbU8`, `tbUd`, `tbKp`, `tbZ16`	signed-permutation test; $2\sqrt{2}U$; $U^\dagger$; `KroneckerProduct`; the 16×16 zero matrix
`tbPs`, `tbEven`	$P_s = \frac{1}{2}(1_4 + s\rho_{4567})$; the even permutations $(4,5,6,7)$, $(4,6,7,5)$, $(4,7,5,6)$
`tbu`, `tbPsi`, `tbBil`	the brane spinor $u = (1, 2, 0, -1)$; $\psi = U(u \otimes e_1)$; $\bar{\psi}\Gamma^\mu\psi$
`tbfW`, `tbu0` … `tbu3`	symbol f_W; the components of $h^{\mu\nu} = \mathrm{diag}(u_0, -u_1, -u_2, -u_3)$

name	meaning
tbAa, tbBb, tbr0, tbth	arguments a, b of a comoving energy $W(a,b)$; a radiation constant; the angle θ
tbT, tbM8, tbTstar, tby	symbols T, M_8; $T_* = (M_8^6/M_{\rm Pl})^{1/5}$; the transverse distance $\lvert\vec{y}\rvert$

14.2 Names of the earlier sections that section 25 uses

name	section	meaning
gCos, a3[x0], b4[x0], cvCos	11.2	the cosmological metric diag$(1,-a^2,-a^2,-a^2,-b^2,\dots,-b^2)$ and its curvature
gPoly, nP, aP, bP, cvPoly	11.3	the inhomogeneous metric diag$(n^2,-a^2,-a^2,-a^2,-b^2,\dots)$ and its curvature
Γ17[A], Γ17up[A], η17, ID16, ID4	3–5	Dirac matrices of 7+1, raised; η; unit matrices
g4D, g5D, rhoK, rBr, uInt, rho4567, PL17	17.2	γ_μ, γ_5, ρ_k, R_A, U, $\rho_4\rho_5\rho_6\rho_7$, the left-handed projector
feStress, bS	20.2	(ε, p_a, p_b) from $W(a,b)$; the symbol b
rIIr4, rIIsig, rIIkmax	22.2	$r_4(k)$, $\sigma(k)$ for $k \le k_{\max}$
rIIVp, rIIM8, rIIMPl	22.2	$V_\perp = M_{\rm Pl}^2/M_8^6$, M_8, $M_{\rm Pl}$
mHb	17.3	$m_h = 125.20$ GeV
dkRhoL	24.5	$\rho_\Lambda = 3H_0^2 M_{\rm Pl}^2 \Omega_\Lambda$